Research Design

Chapter 3

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What is Research Design?

- **Research Design** is defined as a framework or blueprint for conducting the market research project in which are specified the details of the procedures necessary for obtaining the required information to structure and solve the problem.
- Based on a broad approach to the management and research, the research design specifies the details, i.e. the practical aspects of implementing that approach. A good research design will ensure that the market research project is conducted effectively and efficiently.

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What Decision Makers Seek

- Accuracy – i.e. a valid representation of the phenomena under investigation which has come from a reliable or consistent form of measurement and which is sensitive to the important differences in individuals being measured.
- Be current – i.e. as up-to-date as possible.
- Sufficiency – i.e. a complete 'picture' which reflects the characteristics of the market phenomena under study.
- Availability – i.e. access can be made when a decision is imminent.
- Relevance – i.e. the support given makes sense to decision-makers and builds on their existing knowledge.

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Why is This Difficult??

1. The subject of investigation is usually human.

2. The process of measuring or observing humans may cause them to change.

3. It is difficult to assess the effect of extraneous variables in marketing experiments and thus their applications are limited.

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What Should We Prioritize?

- Without **Relevance** ,the whole rationale of supporting the marketing decision maker would be removed

Relevance Cannot Cannot Be Comprimsed

- Relevance permits the decision maker to use the information

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The Participants Perspective

- Getting good answers out of people is difficult
- Some methods require to participants to be more open

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Example: Durex

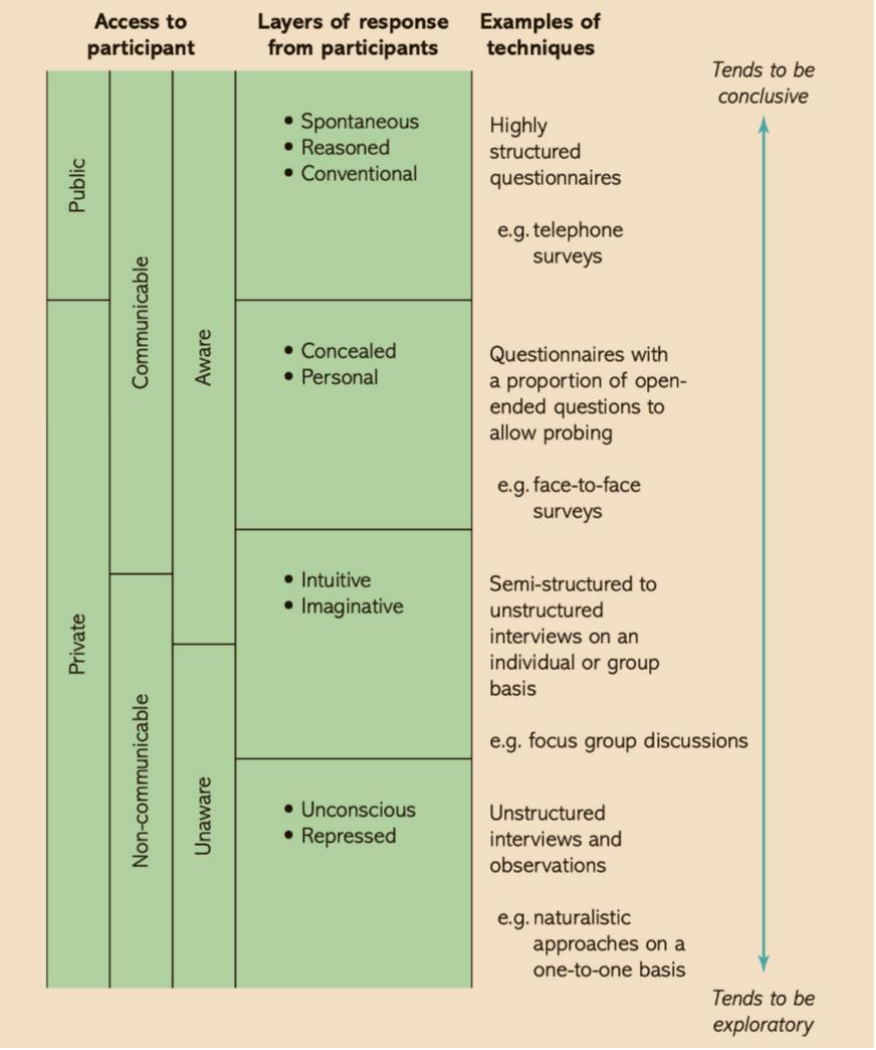
- Durex is a company that produces sexual wellness and intimacy products
- Due to the nature of their products any questions required for their research are considered taboo

How they did the research:

- They were clear about the topics prior to participation
- They used a funnel survey to create trust
- Did it online

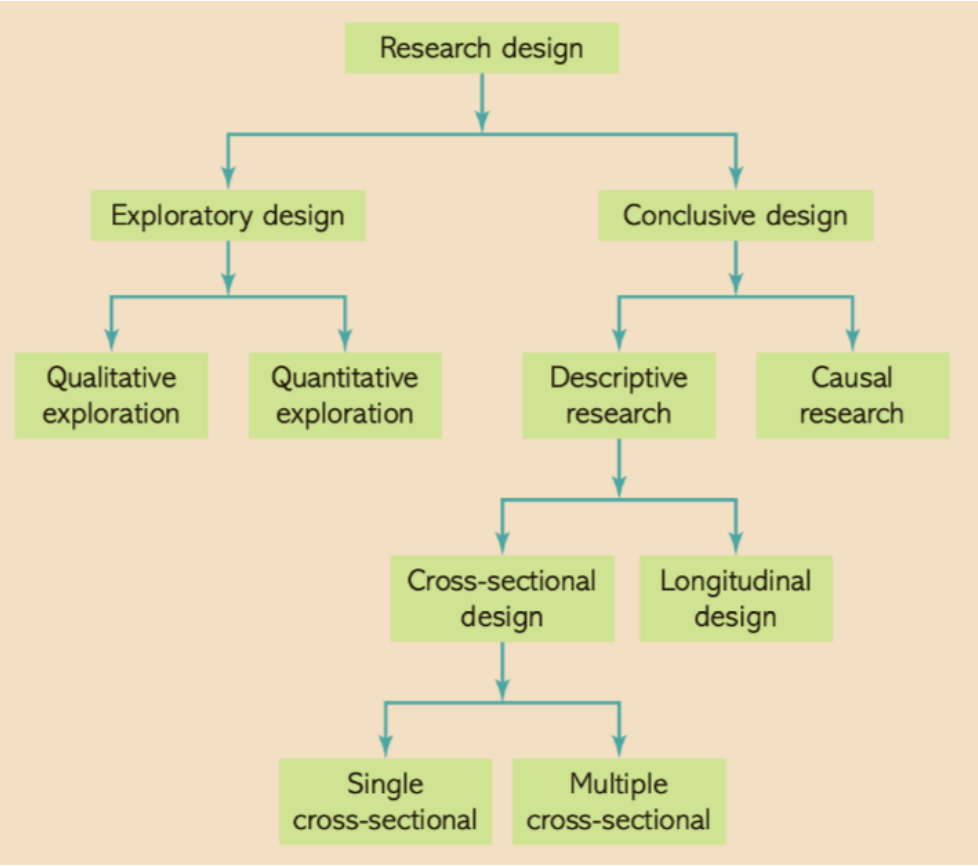
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Responses to Interviews



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Research Design Classification



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Differences Between Exploratory and Conclusive Research

Table 3.1 Differences between exploratory and conclusive research

	Exploratory	Conclusive
Objectives	To provide insights and understanding of the nature of marketing phenomena To understand	To test specific hypotheses and examine relationships To measure
Characteristics	Information needed may be loosely defined Research process is flexible, unstructured and may evolve Samples are small Data analysis can be qualitative or quantitative	Information needed is clearly defined Research process is formal and structured Sample is large and aims to be representative Data analysis is quantitative
Findings/results	Can be used in their own right May feed into conclusive research May illuminate specific conclusive findings	Can be used in their own right May feed into exploratory research May set a context to exploratory findings
Methods	Expert surveys Pilot surveys Secondary data Qualitative interviews Unstructured observations Quantitative exploratory multivariate methods	Surveys Secondary data Databases Panels Structured observations Experiments

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Uses of Exploratory Research Design

1. To obtain some background information where absolutely nothing is known about the problem area
2. To define problem areas fully and to formulate hypotheses for further investigation and/or quantification
3. To identify and explore concepts in the development of new products or forms of marketing communications
4. During a preliminary screening process, such as in new-product development, in order to reduce a large number of possible projects to a smaller number of probable ones
5. To identify relevant or salient behaviour patterns, beliefs, opinions, attitudes, motivations, etc., and to develop structures of these constructs
6. To develop an understanding of the structure of beliefs and attitudes in order to aid the interpretation of data structures in multivariate data analyses
7. To explore the reasons that lie behind the statistical differences between groups that may emerge from secondary data or surveys
8. To explore sensitive or personally embarrassing issues from the participants' and/or the interviewer's perspective
9. To explore issues that participants may feel deeply about, that are difficult for them to rationalise and that they may find difficult to articulate
10. To 'data-mine' or explore quantitative data to reveal hitherto unknown connections between different measured variables

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Uses of Exploratory Research Design

- Background information – obtain background information where absolutely nothing is known about the problem area.
- Formulate hypotheses – for further investigation and/or quantification.
- Concept identification – explore concepts when developing new products or forms of marketing communications.
- Screening – reduce a large number of possible projects into a smaller number of probable ones, such as in new-product development.
- Salient behaviour and attitude patterns – develop structures of these constructs.
- Belief and attitude structures – develop an understanding of these to assist with the interpretation of data structures in multivariate data analyses.
- Explore statistical differences – investigate the reasons which lie behind statistical differences between groups which may emerge from secondary data or surveys.
- Sensitive issues – explore sensitive or personally embarrassing issues from the perspective of the participant and/or interviewer.
- Difficult to articulate – explore issues which are difficult to rationalise and/or articulate.
- Data mine – explore quantitative data to reveal unknown connections between different measured variables.

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Uses of Conclusive Research Design

1. To describe the characteristics of relevant groups, such as consumers, salespeople, organisations, or target market
2. To estimate the percentage in a specified population exhibiting a certain form of behaviour
3. To count the frequency of events, especially in the patterns of consumer behaviour
4. To measure marketing phenomena to represent larger populations or target markets
5. To be able to integrate findings from different sources in a consistent manner, especially in the use of marketing information systems and decision support systems
6. To determine the perceptions of product or service characteristics
7. To compare findings over time that allow changes in the phenomena to be measured
8. To measure marketing phenomena in a consistent and universal manner
9. To determine the degree to which marketing variables are associated
10. To make specific predictions

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Descriptive Research

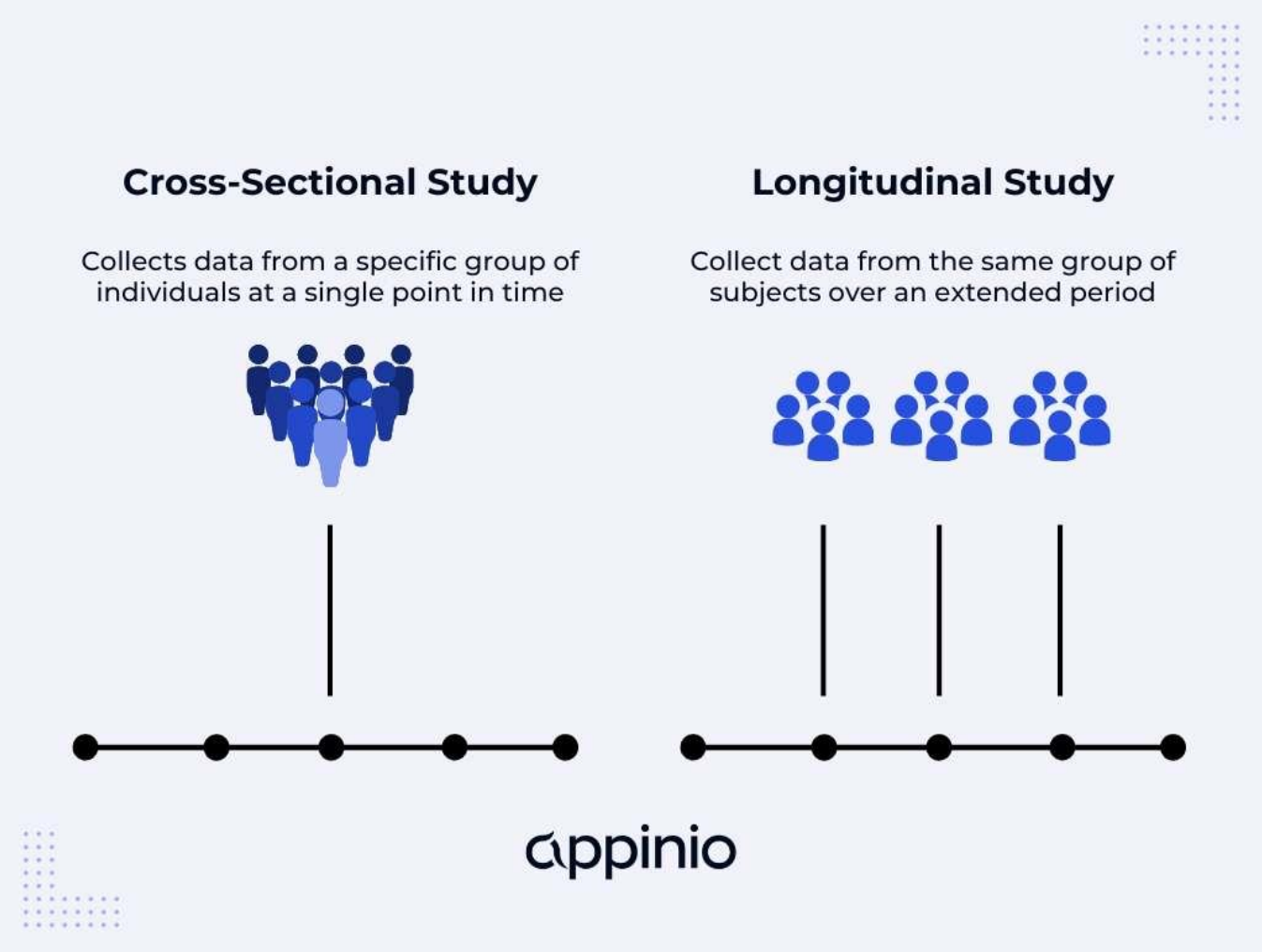
- The major objective is to … describe

Examples:

- Market studies describing the size of the market, buying power of the consumers, availability of distributors and consumer profiles.
- Pricing studies describing the range and frequency of price changes and probable consumer response to proposed price changes
- Image studies determining consumer perceptions of the firm and its products.
- Product usage studies describing consumption patterns

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Two Types of Study



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Cross-Sectional Design

Single Cross-Sectional

- only one sample of participants is drawn from the target population
- information is obtained from this sample only once

Multiple Cross-Sectional

- there are two or more samples of participants, and information from each sample is obtained only once
- Often, information from different samples is obtained at different times.

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Cohort Analysis

- **Cohort Analysis** consists of a series of surveys conducted at appropriate time intervals, where the cohort serves as the basic unit of analysis.

- A cohort is a group of participants who experience the same event within the same time interval. (Birth Cohorts)

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Consumption of Soft Drinks by Various Age Cohorts

(Percentage consuming on a typical Day)

Age	1950	1960	1970	1980	
8–19	53	63	73	81↘	
20–29	45	61	76	76↘	C8
30–39	34	47	68	71↘	C7
40–49	23	41	59	68↘	C6
50+	18	29	50	52↘	C5
		C1	C2	C3	C4

C1: cohort born prior to 1900
C2: cohort born 1901–1910
C3: cohort born 1911–1920
C4: cohort born 1921–1930
C5: cohort born 1931–1940
C6: cohort born 1941–1950
C7: cohort born 1951–1960
C8: cohort born 1961–1970

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Longitudinal Designs

- In a **longitudinal design**, a fixed sample (or samples) of population elements is measured repeatedly

Example: Why do you think companies have "loyalty" rewards programs?

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Boring But Necessary Definitions

- **Panel-** A sample of participants who have agreed to provide information at specified intervals over an extended period.

- **Access Panel-** A general ‘pool’ of individuals or households who have agreed to be available for surveys of widely varying types and topics.

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Comparisons of Designs

Evaluation criteria	Cross-sectional design	Longitudinal design
Detecting change	–	+
Large amount of data collection	–	+
Accuracy	–	+
Representative sampling	+	–
Response bias	+	–

Note: + indicates a relative advantage over the other design, whereas – indicates a relative disadvantage.

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Detecting Change

Cross-Sectional Data

Brand purchased	Time period	
	Period 1 survey	Period 2 survey
Total surveyed	1,000	1,000
Brand A	200	200
Brand B	300	300
Brand C	500	500

Longitudinal Data

Brand purchased in period 1	Brand purchased in period 2			
	Brand A	Brand B	Brand C	Total
Total surveyed	200	300	500	1,000
Brand A	100	50	50	200
Brand B	25	100	175	300
Brand C	75	150	275	500

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Causal Research

- A type of conclusive research where the major objective is to obtain evidence regarding cause-and-effect (causal) relationships.

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Causal Research

1. To understand which variables are the cause (independent variables) and which variables are the effect (dependent variables) of marketing phenomena.
2. To determine the nature of the relationship between the causal variables and the effect to be predicted.
3. To test hypotheses.

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Challenge of Causal Research

- causal or independent variables must be manipulated in a relatively controlled environment
- The environment is one in which the other variables that may affect the dependent variable are controlled or checked as much as possible.
- The effect of this manipulation on one or more dependent variables is then measured to infer causality.

The main method of causal research is experimentation.

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Exploratory Descriptive Causal

When little is known about the problem situation, it is desirable to begin with **exploratory research**

Exploratory research is appropriate for the following:

- When the nature of the topic under study cannot be measured in a structured, quantifiable manner.
- When the problem needs to be defined more precisely.
- When alternative courses of action need to be identified.
- When research questions or hypotheses need to be developed.
- When key variables need to be isolated and classified as dependent or independent.

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Exploratory Descriptive Causal

- Exploratory research can be the starting point to move towards a better descriptive or causal research

- Exploratory may be skipped or follow descriptive or causal if it provides better insights to help understand the findings

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Error in Research

- Several potential sources of error can affect a research design.

- A good research design attempts to control the various sources of error

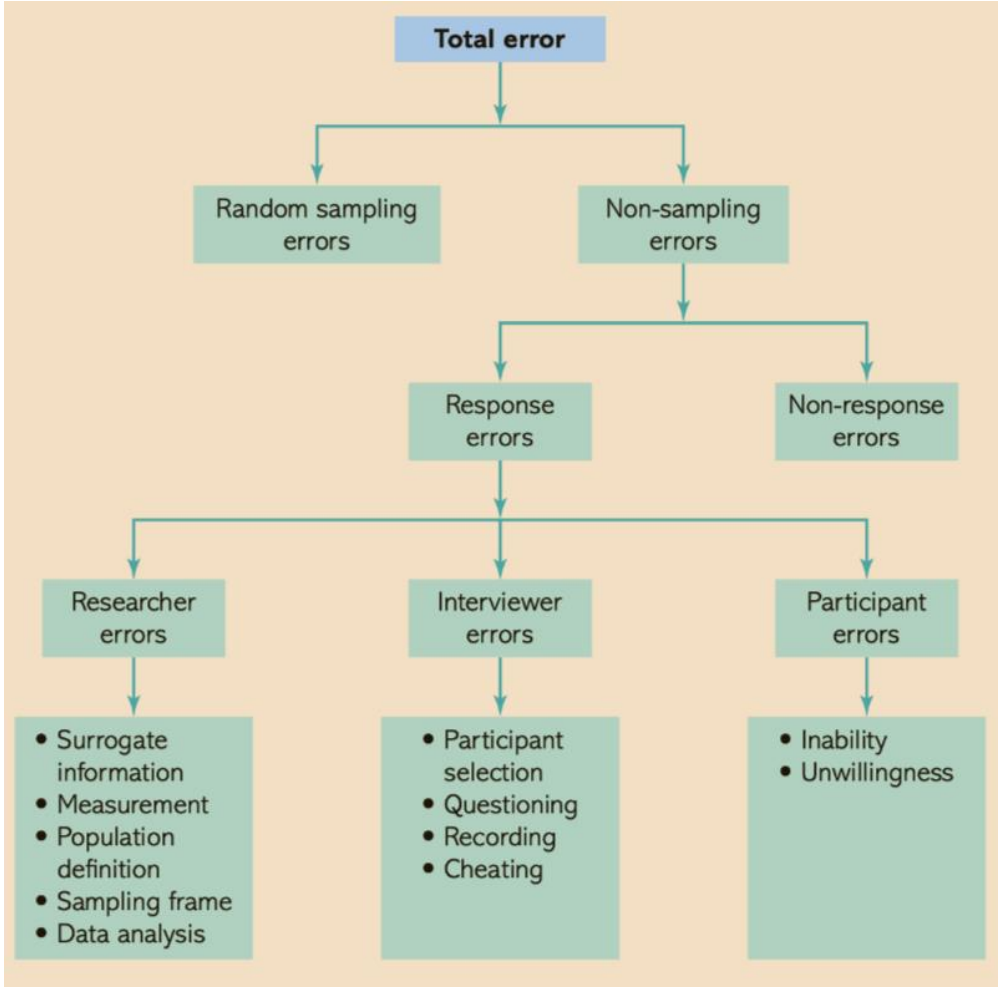
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Total Error

- Total Error is The variation between the true mean value in the population of the variable of interest and the observed mean value obtained in the marketing research project.

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Potential Sources of Error in Research Designs



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Separating Error

Random Sampling Error

- The error arising because the particular sample selected is an imperfect representation of the population of interest
- It may be defined as the variation between the true mean value for the sample and the true mean value of the population.

Non-Sampling Error

- An error that can be attributed to sources other than sampling and that can be random or nonrandom.

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Separating Non-Sampling Error

Response Error

- A type of non-sampling error arising from participants who do respond but who give inaccurate answers or whose answers are mis-recorded or mis-analyzed.
- defined as a variation between the true mean value of the variable in the net sample and the observed mean value obtained in the market research project.

Non-Response Error

- A type of non-sampling error that occurs when some of the participants included in the sample do not respond.
- This error may be defined as the variation between the true mean value of the variable in the **original** sample and the true mean value in the **net sample.**

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Researcher Error

- Surrogate Information Error
  - Measurement Error
- Population Definition Error
  - Sampling Frame Error
  - Data Analysis Error

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Surrogate Information Error

- **Surrogate information error** may be defined as the variation between the information needed for the marketing research problem and the information sought by the researcher.

- instead of obtaining information on consumer choice of a new brand (needed for the marketing research problem), the researcher obtains information on consumer preferences because the choice process cannot be easily observed.

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Measurement Error

- Measurement error may be defined as the variation between the information sought and information generated by the measurement process employed by the researcher.

- While seeking to measure consumer preferences, the researcher employs a scale that measures perceptions rather than preferences.

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Population Definition Error

- Population definition error may be defined as the variation between the actual population relevant to the problem at hand and the population as defined by the researcher. The problem of appropriately defining the population may be far from trivial

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Population Definition Error Example

The population of the affluent households was defined in four different ways in a study:

1. Households with an income of €80,000 or more.
2. The top 20% of households, as measured by income.
3. Households with net worth over €450,000.
4. Households with discretionary income to spend being 30% higher than that of comparable households

Their number and characteristics varied depending on the definition, underscoring the need to avoid population definition error.

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Sampling Frame Error

- Sampling frame error may be defined as the variation between the population defined by the researcher and the population as implied by the sampling frame (list) used.

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Sampling Frame Error Example

- The telephone directory used to generate a list of telephone numbers does not accurately represent the population of potential landline consumers due to unlisted, disconnected and new numbers in service. It also misses out the great number of consumers who choose not to have landlines, exclusively using mobile phones.

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Data Analysis Error

- **Data analysis error** encompasses errors that occur while raw data from questionnaires are transformed into research findings.

- For example, an inappropriate statistical procedure is used, resulting in incorrect interpretation and findings.

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Interviewer Error

- **Participant Selection Error** occurs when interviewers select participants other than those specified by the sampling design, or in a manner inconsistent with the sampling design.

- **Questioning Errors** denotes errors made when asking questions of the participants, or in not probing when more information is needed. For example, while asking questions an interviewer does not use the exact wording or prompts as set out in the questionnaire.

- **Recording Error** arises due to errors in hearing, interpreting and recording the answers given by the participants. For example, a participant indicates a neutral response (undecided) but the interviewer misinterprets that to mean a positive response (would buy the new brand).

- **Cheating Error** arises when the interviewer fabricates answers to a part or the whole of the interview. For example, an interviewer does not ask the sensitive questions related to a participant's debt but later fills in the answers based on personal assessment.

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Participants Error

Inability Error

- results from the participant's inability to provide accurate answers.
- Participants may provide inaccurate answers because of unfamiliarity, fatigue, boredom, question format, question content or because the topic is buried deep in the participant's mind.

An example of inability error is where a participant cannot recall the brand of toothpaste he or she purchased four weeks ago.

Unwillingness Error

- arises from the participant's unwillingness to provide accurate information.
- Participants may intentionally misreport their answers because of a desire to provide socially acceptable answers, because they cannot see the relevance of the survey and/or a question posed, to avoid embarrassment or to please the interviewer

For example, to impress the interviewer a participant intentionally says that he or she reads *The Economist* magazine

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How to Eliminate Error

- the researcher should attempt to minimize the total error, not just a particular source

- Ex: Increasing the sample size does decrease sampling error, but it may also increase non-sampling error, e.g. by increasing interviewer errors
- Sampling error can be calculated, whereas many forms of non-sampling error defy estimation.

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Questions

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